Tutorial 2 — Multiple Regression and Partial Effects

Introductory Econometrics — Télécom Paris

Theory exercise

Exercise 1 (What does a higher R^2 establish?)

Using the same n observations, consider the nested OLS models

$$A: \quad y_i = a_0 + a_1 x_{1i} + u_i,$$
$$B: \quad y_i = b_0 + b_1 x_{1i} + b_2 x_{2i} + v_i.$$

Let RSS_A and RSS_B denote their residual sums of squares, and assume that both models contain an intercept and have full column rank.

(a) Use the fact that OLS minimizes the residual sum of squares to show that $RSS_B \leq RSS_A$.

(b) Using $R^2 = 1 - RSS/TSS$, show that $R^2_B \geq R^2_A$. When will the two values be equal?

(c) A researcher concludes that Model B is necessarily a better causal model because it has the larger R^2. Explain why this conclusion is wrong.

(d) Explain why adjusted R^2 can decrease when x_2 is added. Derive the condition under which Model B has at least as large an adjusted R^2 as Model A.

Python lab

Use `code/lab2.py` for all Python work. Keep the supplied `code/` and `data/` folders together. The script locates the original `data/WAGE1.raw` file automatically and creates `code/output/`. Guided blanks are marked `TODO`. Lab 3 contains complete Monte Carlo code so that class time can focus on its econometric meaning.

Lab 1. Does adding controls solve the education–wage problem?

Study the following question:

> How does the education–wage association change when experience and tenure are included, and does the resulting multiple regression identify a causal return to education?

The data contain 526 workers from a historical US cross-section based on the 1976 Current Population Survey. `wage` is hourly earnings in 1976 US dollars; `educ`, `exper`, and `tenure` are measured in years.

Step 1. Define the comparison before estimating it

The script loads the common estimation sample and reports the correlations among education, experience, and tenure.

(a) In Model B, which workers are compared when we interpret the education coefficient? What is held fixed, and why is this not yet a causal interpretation?

(b) Education is negatively correlated with experience and tenure in this sample. If experience and tenure have positive partial associations with wages, would you expect omitting them to push the simple education coefficient upward or downward? Why is this sign check suggestive rather than conclusive?

Step 2. Estimate and explain the coefficient change

Estimate on the same sample

$$A: \quad \log(wage_i) = a_0 + a_1 educ_i + v_i,$$
$$B: \quad \log(wage_i) = b_0 + b_1 educ_i + b_2 exper_i + b_3 tenure_i + u_i.$$

(a) How much does the education coefficient change between Models A and B? Interpret Model B's coefficient in one sentence, using an approximate percentage and stating what is held fixed.

(b) Does the direction of the change match your prediction in Step 1? Explain what happens to experience and tenure when Model A omits them.

(c) Why was Model B's R^2 guaranteed not to be smaller than Model A's? Does the observed increase make zero conditional mean more credible?

Step 3. Audit exogeneity as part of the analysis

(a) In practical terms, what would $\mathbb{E}(u \mid educ, exper, tenure) = 0$ require? Identify three plausible wage determinants left in u, explain why each may be related to education, and say which one concerns you most.

(b) Suppose the dataset also contained parents' education, a test score measured before schooling, and current occupation. Which variables might help proxy for pre-existing confounders, and which might lie on a pathway from education to wages?

(c) Why is "add every available variable" a poor rule for choosing controls? Give two reasons, considering the timing of experience, tenure, and occupation as well as the possibility of selection.

(d) Write three concise sentences: report the conditional education association; explain why the coefficient changes relative to Model A; and state the principal remaining threat to a causal interpretation.

Lab 2. What partialling out does—and does not—show

Use experience and tenure as the controls.

(a) Regress education on the controls, including an intercept, and save r_x. What does a positive education residual mean?

(b) Following Lecture 2, slide 14, calculate

$$\frac{\sum_i r_{xi} \log(wage_i)}{\sum_i r_{xi}^2}.$$

Verify that it equals Model B's education coefficient. What variation in education identifies this coefficient?

(c) Now regress log wage on the controls, including an intercept, and save the residual r_y. Regress r_y on r_x with an intercept. Verify that the intercept is numerically zero and that the slope equals the result in part (b). Explain why the two procedures are equivalent. Does their equality provide evidence that $\mathbb{E}(u \mid educ, exper, tenure) = 0$?

Lab 3. When a precise regression answers the wrong question

We now use a Monte Carlo experiment to separate bias from imprecision. The Python script repeatedly draws samples of $n = 150$ observations from

$$y = 1 + x_1 + x_2 + u, \qquad \mathbb{E}(u \mid x_1, x_2) = 0,$$

first with $\mathrm{Corr}(x_1, x_2) = 0.2$ and then with $\mathrm{Corr}(x_1, x_2) = 0.99$. In every sample it estimates the correct full model and the short model that omits x_2. Repeating this process approximates the sampling distributions of the coefficient estimates. The simulation code is already included so that the analysis can focus on comparing those distributions.

(a) Before running the code, predict the centres of the full-regression coefficient on x_1 and the short-regression slope. For the short regression, use the lecture's omitted-variable result and note that the slope from regressing x_2 on x_1 is ρ here.

(b) Run the simulation. How do the mean and sampling standard deviation of the full-regression coefficient change as correlation rises? Is high multicollinearity creating bias?

(c) Explain the strong negative relationship between the two full-model coefficient estimates in the high-correlation design. Why is their sum much more stable?

(d) Compare the full and short estimates when $\rho = 0.99$. The short slope has a much smaller sampling standard deviation. Why is it nevertheless the wrong estimator of the partial effect of x_1?

(e) A colleague proposes deleting x_2 because it makes the standard error on x_1 large. What would you tell them?